

Project Proposal

Project Title:

AI-Powered Gym Management System

Project Description:

The AI-Powered Gym Management System is a web-based application designed to simplify gym operations and improve the fitness experience for members. The system will manage member registration, attendance, membership plan, trainer assignments, and workout records. An AI-based recommendation feature will suggest personalized workout plan based on users fitness goals, age, BMI, and experience level.

Problem Statement:

Many small gyms still rely on manual record keeping, making it difficult to manage memberships, attendance, and personalized fitness guidance. This project aims to digitalized gym management while introducing basic AI to recommend suitable workout plans for members.

Objectives:

- Develop a user-friendly Gym Management System.
- Store and manage member information using a relational database.
- Implement an AI-based workout recommendation feature.
- Track attendance and membership details.
- Generate reports for gym administrators.

Proposed Features:

- Member Registration and Login
- Membership Management
- Attendance Tracking
- Trainer Management
- Workout history
- AI workout Recommendation
- Admin Dashboard
- Reports and statistics

Technologies:

- Frontend: HTML, CSS, JavaScript
- Backend: Python (Flask)
- Database: MySQL
- AI: Python (Rule-Based Recommendation or Decision Tree)
- Version Control: GitHub

Expected Outcome:

The project will provide an efficient Gym Management Solution with basic AI capabilities for personalized workout recommendations. It will show the integration of Artificial Intelligence with database management while solving a real world problem.

Team Members and Responsibilities:

Members Name	Responsibilities
Aayush Shah	Frontend Development
Riwaj Khadka	Backend Development
Nauraj Khawas & Ankit Das	Database Design and Implementation
Darpan Subedi	Testing, Documentation and Presentation
Aayush Shah	AI Recommendation Module